REGULATORY DOSSIER SUBMISSION
Dossier ID: DOS-E9C22FF76D
Country: Botswana
Submission Date: 2025-07-29
Product: Propmed (propranolol) | ATC: C07AA05 | Strength: 80 mg
Applicant: Alpha Therapeutics | Manufacturer: Dodoma Pharma Works
Policy Label: standard_review | Risk Score: 0.5
AMR Stewardship: AWaRe=not_applicable | Unmet Need=not_applicable | GLASS Trend=not_applicable | Watch Similarity=not_applicable

SECTION CONTENT

[1] m1_application_admin - Application Form and Administrative Information
Labels: presence=present; length=length_ok; correctness=correct
This dossier constitutes a formal application for Marketing Authorization of Propmed (propranolol) submitted on 2025-07-29 to the national regulatory authority of Botswana. The applicant, Alpha Therapeutics, formally requests approval for the commercial distribution of this medicinal product. The proposed pharmaceutical form is tablet with a strength of 80 mg, classified under ATC code C07AA05. All required administrative declarations, legal attestations, letters of authorization, and regional application forms have been completed, signed by the responsible Qualified Person, and appended to Module 1. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Regional administrative forms were checked for signature completeness and dossier version alignment. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. All referenced documents are available in the dossier annex and traceable by section identifier. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Administrative cover letters, declarations, and fee records were cross-checked against the submission index. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The proposed prescribing information and patient informatio

[1] m1_manufacturer_gmp - Manufacturer and GMP Evidence
Labels: presence=present; length=length_ok; correctness=correct
The primary commercial manufacturing site for the finished pharmaceutical product is Dodoma Pharma Works, located in Tanzania. The facility was subject to a comprehensive GMP inspection by a recognized stringent regulatory authority. The latest inspection concluded on 2025-09-03, resulting in a formal compliance status of ’compliant’. The active GMP certificate number is GMP-260265, which remains valid until 2027-03-27. Site master files, a summary of recent inspection observations, and evidence of completed Corrective and Preventive Actions (CAPA) are included in the annex. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Recent regulatory inspections yielded no critical observations, and all major findings have been effectively closed. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Site quality-system commitments were compared against the

proposed commercial manufacturing activities. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Inspection history, CAPA closure records, and site responsibilities were reconciled across the manufacturing chain. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Method validation records and quality controls were reviewed for internal consistency. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. All refe

Site Inspection History Summary:

Inspection Date	Authority	Scope	Outcome
2025-09-03	National Auth	FPP/General	compliant
2022-05-12	SRA Joint	API/Sterile	compliant

[1] m1_product_information - Product Information and Naming
Labels: presence=present; length=length_ok; correctness=correct
The proposed proprietary name for this medicinal product is Propmed, containing the active pharmaceutical ingredient propranolol. The draft Summary of Product Characteristics (SmPC), Patient Information Leaflet (PIL), and primary/secondary packaging labels are provided for review. These documents detail the approved therapeutic indications, posology, contraindications, special warnings, and precautions for use. A comprehensive risk management plan to minimize medication errors is also submitted. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The applicant confirms alignment with applicable technical guidance and regional submission standards. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribu

[2] m2_quality_overall_summary - Quality Overall Summary
Labels: presence=present; length=length_ok; correctness=correct
This Quality Overall Summary (QOS) provides a critical evaluation of the chemistry, manufacturing, and controls (CMC) data for propranolol. The control strategy justifies the proposed specifications for both the active substance and the finished product, emphasizing critical quality attributes (CQAs) and critical process parameters (CPPs). Impurity profiles, including degradation products and potential genotoxic impurities, have been thoroughly characterized. Batch analysis data from three commercial-scale validation batches demonstrate consistent manufacturing performance and compliance with all release criteria. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Method validation records and quality controls were reviewed for internal consistency. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. All referenced documents are available in the dossier annex and traceable by section identifier. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The applicant confirms alignment with applicable technical guidance and regional submission standards. Data were analyzed using a mixed-effects

model for repeated measures, adjusting for baseline covariates. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The risk mana

[2] m2_clinical_overview - Clinical Overview and Benefit-Risk Summary
Labels: presence=present; length=length_ok; correctness=correct
The Clinical Overview synthesizes the efficacy and safety data supporting the use of Propmed for the treatment of community acquired pneumonia. The pivotal clinical development program comprised 4 well-controlled studies. Based on the integrated analysis, the reported clinical outcome category is 'endpoint_met'. The overall benefit-risk profile is considered highly favorable for the target patient population. Safety findings were consistent with the known pharmacological class effects, and appropriate risk minimization measures have been integrated into the proposed prescribing information. Benefit-risk language was reviewed for consistency with the submitted efficacy and safety summaries. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The target population demographics are broadly representative of the intended commercial patient cohort. The applicant confirms alignment with applicable technical guidance and regional submission standards. Method validation records and quality contro

[3] m3_api_quality - API Quality and Control Strategy
Labels: presence=present; length=length_ok; correctness=correct
The active pharmaceutical ingredient (API) section details the complete synthesis route, starting from well-defined regulatory starting materials. Robust in-process controls and intermediate specifications ensure the consistent quality of the final drug substance. The analytical procedures used for release and stability testing have been fully validated for accuracy, precision, linearity, and robustness. The impurity control strategy adequately addresses organic impurities, residual solvents, and elemental impurities in accordance with current ICH guidelines. Forced degradation studies confirm the stability-indicating nature of the assay methods. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. All referenced documents are available in the dossier annex and traceable by section identifier. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Residual solvents and elemental impurities are controlled well below ICH Q3C and Q3D permissible daily exposures. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The synthetic route involves three well-characterized steps with defined starting materials and isolated intermediates. Method validation records and quality controls were reviewed for internal consistency. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The applicant confirms alignment with applicable technical

guidance and regional submission standards. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The API package links specification limits to validated analytical methods and impurity-control decisions. Critical material attributes and control strategy elements were reviewed for consistency across the quality dossier. Batch analysis, validation, and impurity management records sup

[3] m3_fpp_manufacturing - FPP Manufacturing Process and Controls
Labels: presence=present; length=length_ok; correctness=correct
The finished pharmaceutical product (FPP) manufacturing process utilizes standard, scalable unit operations. A formal quality risk assessment (e.g., FMEA) was conducted to identify critical process parameters (CPPs) that impact critical quality attributes (CQAs). Process performance qualification (PPQ) reports for three consecutive commercial-scale batches verify that the process is maintained in a state of control. In-process controls, intermediate hold-time studies, and packaging validation data are presented to substantiate the robustness and reproducibility of the commercial manufacturing operations. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. All referenced documents are available in the dossier annex and traceable by section identifier. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstr

[3] m3_stability - Stability and Shelf-Life Justification
Labels: presence=present; length=length_ok; correctness=correct
A comprehensive stability program has been executed to justify the proposed shelf-life and storage conditions. Data from both accelerated (40℃/75% RH for 6 months) and long-term storage conditions (spanning up to 36 months) are presented for multiple primary stability batches. Statistical evaluation using linear regression and poolability tests confirms that degradation trends remain well within the acceptable specification limits. Photostability and freeze-thaw cycling studies demonstrate that the product does not require specific handling precautions. A post-approval stability protocol commitment is included. The applicant confirms alignment with applicable technical guidance and regional submission standards. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant wi

[4] m4_nonclinical_summary - Nonclinical Study Summary
Labels: presence=present; length=length_ok; correctness=correct
The nonclinical testing program evaluated the primary pharmacodynamics, secondary pharmacodynamics, safety pharmacology, and comprehensive toxicology profile of the compound. Repeat-dose toxicity studies in two mammalian species established clear no-observed-adverse-effect levels (NOAELs). Genotoxicity testing (Ames and in vivo micronucleus assays) yielded negative results. Reproductive and developmental toxicity studies revealed no teratogenic signals at clinically relevant exposures. The calculated safety margins support the proposed clinical dosing regimen without major safety concerns. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The applicant confirms alignment with applicable technical guidance and regional submission standards. All referenced documents

[5] m5_trial_listing - Tabular Listing of Clinical Studies
Labels: presence=present; length=length_ok; correctness=correct
The tabular listing of clinical studies provides a comprehensive inventory of all trials conducted in support of the proposed indication (community acquired pneumonia). This includes Phase I pharmacokinetic/pharmacodynamic studies, Phase II dose-finding studies, and the pivotal Phase III efficacy and safety trials. For each study, the table details the protocol identifier, study design, randomization ratio, number of enrolled subjects, study duration, and current completion status. The applicant confirms alignment with applicable technical guidance and regional submission standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoin

Pivotal Clinical Trial Inventory:

Protocol ID	Phase	Enrollment	Status
PROT-001	III	433	completed
PROT-002	III	264	completed
PROT-003	III	756	completed
PROT-004	III	791	completed

[5] m5_pivotal_trial_reports - Pivotal Clinical Trial Reports
Labels: presence=present; length=length_ok; correctness=correct
Detailed clinical study reports (CSRs) for the pivotal efficacy trials are provided. These randomized, double-blind, actively-controlled trials evaluated the primary and secondary endpoints in accordance with the pre-specified statistical analysis plan (SAP). The trials resulted in a formal outcome category of ’endpoint_met’, demonstrating robust treatment effects in the intent-to-treat (ITT) and per-protocol populations. Subgroup analyses across age, gender, and baseline disease severity strata were consistent with the primary findings. Serious adverse events (SAEs) and discontinuations were systematically analyzed and adjudicated. Method validation records and quality controls were reviewed for internal consistency. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The applicant confirms alignment with applicable technical guidance and regional submission standards. The primary efficacy endpoint was met, demonstrating a statistically significant improvement over placebo ($p < 0.001$). Clinical study reports document protocol adherence, endpoint handling, and safety interpretation in the target population. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The pivotal study

narratives align endpoint definitions, analysis populations, and outcome interpretation. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The trial reports connect efficacy conclusions to the prespecified analysis framework and observed safety findings. Discontinuations due to treatment-emergent adverse events (TEAEs) were low and balanced across all study arms. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. All referenced documents are available in the dossier annex and traceable by section identifier. Method validation records and quality controls were reviewed for internal consistency. The applicant confirms alignment with applicable technical guidance and regional submission standards. The applicant confirms alignment with applicable technical guidance and regional submission standards. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Discontinuations due to treatment-emergent adverse events (TEAEs) were low and balanced acr

[5] m5_bioequivalence - Biopharmaceutics and Bioequivalence Evidence
Labels: presence=present; length=length_ok; correctness=correct
Biopharmaceutics data substantiate the formulation development bridging between clinical trial materials and the proposed commercial product. In vivo bioequivalence studies demonstrated that the 90% confidence intervals for the geometric mean ratios of Cmax and AUC parameters fell entirely within the standard acceptance criteria of 80.00% to 125.00%. Additionally, multi-point comparative dissolution profiles across physiological pH ranges (pH 1.2, 4.5, and 6.8) confirmed similarity ($f_2 > 50$). The bioanalytical methods used for plasma quantification were fully validated. Method validation records and quality controls were reviewed for internal consistency. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The applicant confirms alignment with applicable technical guidance and regional submission standards. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. All referenced documents are available in the dossi